

Propositional logic

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6. Value of a formula

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A little grammar

Formula comes from the Latin language

Plurals:

Formulas: plural by the English method – common speech, e.g.

Instead of exact formulas, one can use asymptotic information.

Formulae: plural by the Latin method – in academic circles where the plural formulae is often the norm, e.g.

This specification is based on the view of objects as atomic formulae.

Propositional logic

Zero order language

An ordered triple $\mathcal{L}^{(0)} = \langle \mathbf{LC}, \mathbf{Con}, \mathbf{Form} \rangle$ defines the language of the zero-order (propositional) logic, where

1. $\mathbf{LC} = \{ \neg, \supset, \wedge, \vee, \equiv, (,) \}$ is the set of *logical constants*,
2. $\mathbf{Con} \neq \emptyset$ is the set of *non-logical constants* such that

$$\mathbf{LC} \cap \mathbf{Con} = \emptyset;$$

3. $\mathbf{Form}$ is the set of *formulae*.

$\neg$	Negation
$\vee$	Disjunction
$\wedge$	Conjunction
$\supset$	Implication
$\equiv$	Equivalency

Formulas in a Zero order language

The words of the zero-order language are the members of the **Form** set, and we call them *zero-order formulas*. The **Form** set is inductively defined as follows:

- **Con** $\subseteq$ **Form** (if **A** $\in$ **Con** then **A** *atomic*)
- if **B** $\in$ **Form** then $\neg$ **B** $\in$ **Form**.
- if **B** $\in$ **Form** and **C** $\in$ **Form**, then
 - **(B $\wedge$ C)** $\in$ **Form**,
 - **(B $\vee$ C)** $\in$ **Form**,
 - **(B $\supset$ C)** $\in$ **Form**,
 - **(B $\equiv$ C)** $\in$ **Form**.

Zero order language

The logical and non-logical constants form the **letters** of the zero-order language

The formulas can be considered as the **words** of the zero-order language

Direct subformulae

If $A \in \text{Con}$ then

- the A atomic formula has no direct subformula;

if $B \in \text{Form}$ and $C \in \text{Form}$ are arbitrary formulas, then

- the formula $\neg B$ has one direct subformula: the formula B ;
- the formulae $(B \wedge C)$, $(B \vee C)$, $(B \supset C)$ and $(B \equiv C)$ have two direct subformulae: the formula B and the formula C .

Set of subformulae

The *set of subformulae* of a formula $A \in \mathbf{Form}$ is set of formulae denoted by $SF(A)$ and is defined inductively as follows:

1. $A \in SF(A)$;
2. if $B \in SF(A)$ and C is a direct subformula of B , then $C \in SF(A)$.

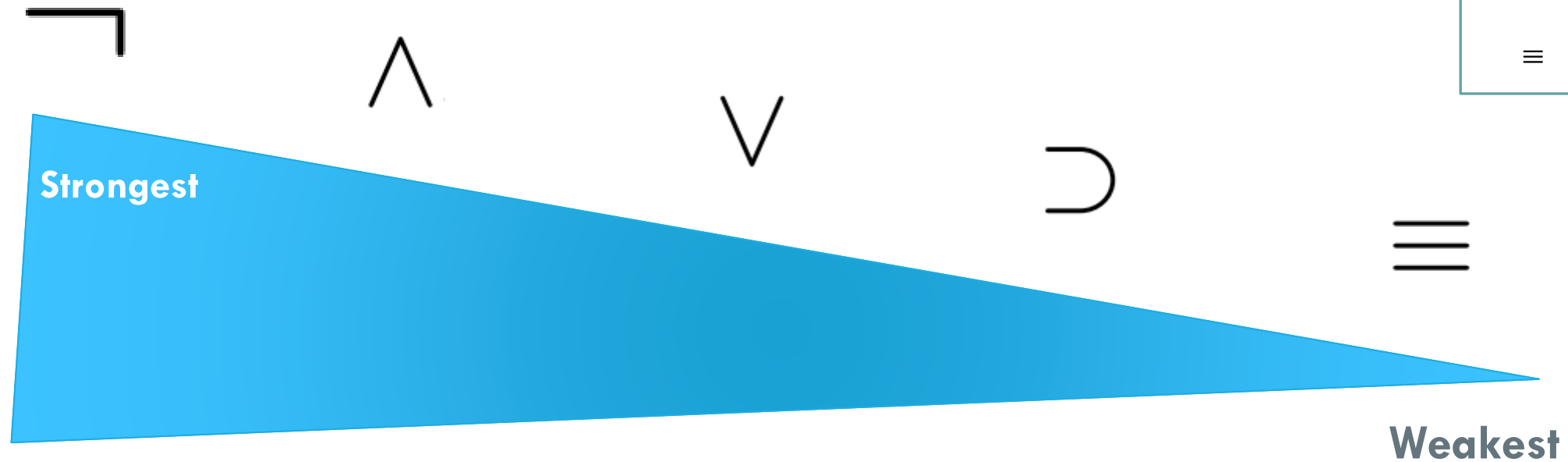
Note that SF is a function such that:

$$SF : \mathbf{Form} \rightarrow 2^{\mathbf{Form}} \setminus \{\emptyset\}$$

Precedence

Connectives:

- $\neg$ Negation
- $\vee$ Disjunction
- $\wedge$ Conjunction
- $\supset$ Implication
- $\equiv$ Equivalency



Parentheses can be dropped when the structure of an expression can be determined by precedence.

Precedence

If surrounded by two occurrences of $\wedge$ or $\vee$, the operand associates with the operator to the left.

$$\begin{aligned}p \wedge q \wedge r &\rightarrow ((p \wedge q) \wedge r) \\p \vee q \vee r &\rightarrow ((p \vee q) \vee r)\end{aligned}$$

If surrounded by two occurrences of $\supset$ or $\equiv$, the operand associates with the operator to the right.

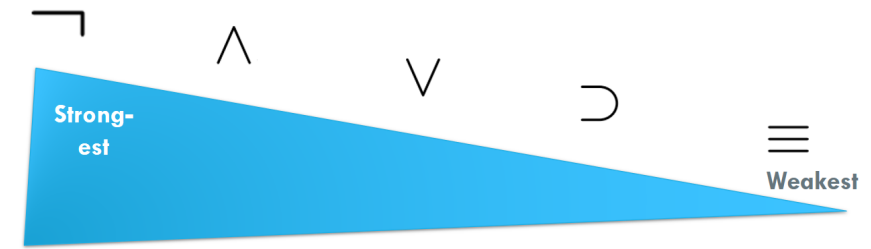
$$\begin{aligned}p \supset q \supset r &\rightarrow (p \supset (q \supset r)) \\p \equiv q \equiv r &\rightarrow (p \equiv (q \equiv r))\end{aligned}$$

Construction tree

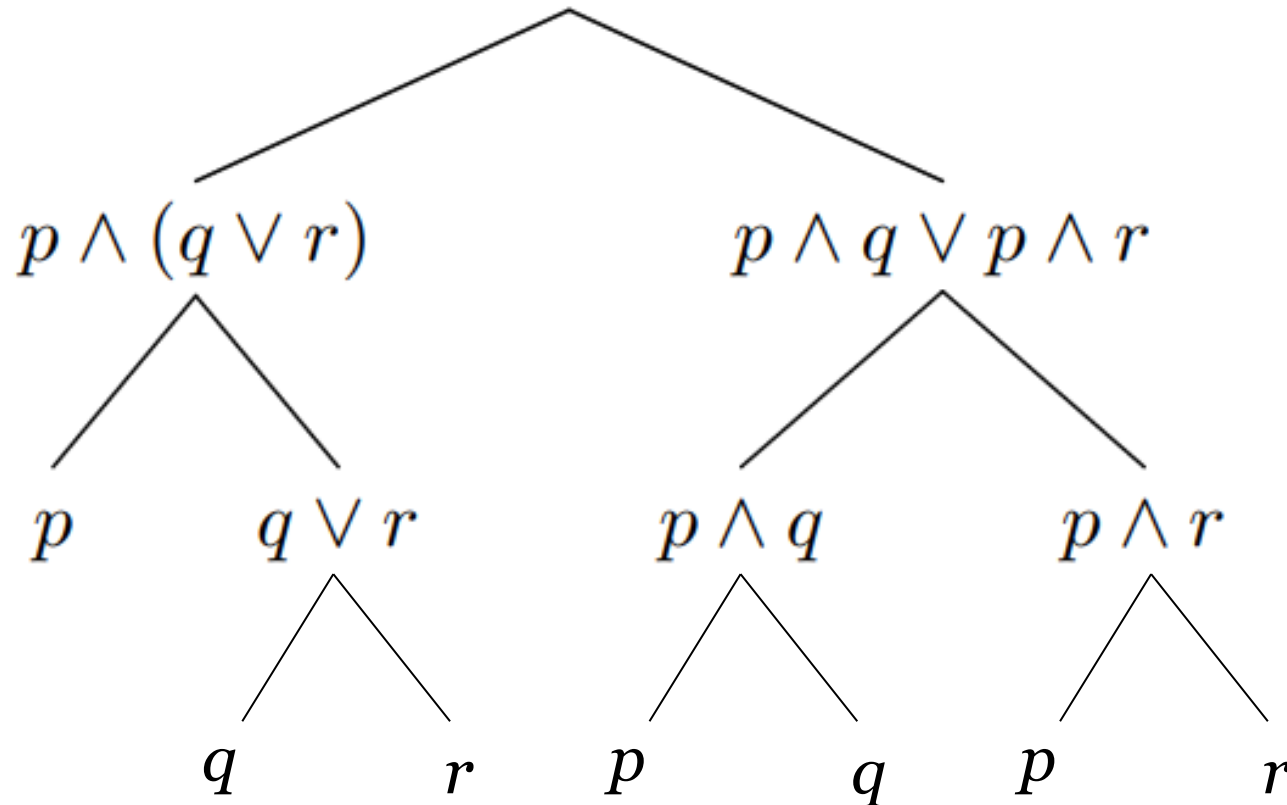
The construction tree of a formula $A \in \text{From}$ is the finite ordered binary tree defined as follows:

- their nodes are formulae,
- their root is labelled by the formula A ,
- a node labelled by $\neg B$ has only one child labelled by B ,
- a node labelled by $(B \wedge C)$, $(B \vee C)$, $(B \supset C)$ or $(B \equiv C)$ has exactly two children labelled by B és C ,
- the leaves labelled by atomic formulae.

Construction tree



$$p \wedge (q \vee r) \equiv p \wedge q \vee p \wedge r$$



You have to find the direct subformulae of the root formula.

Then you have to find the direct subformulae of these subformulae.

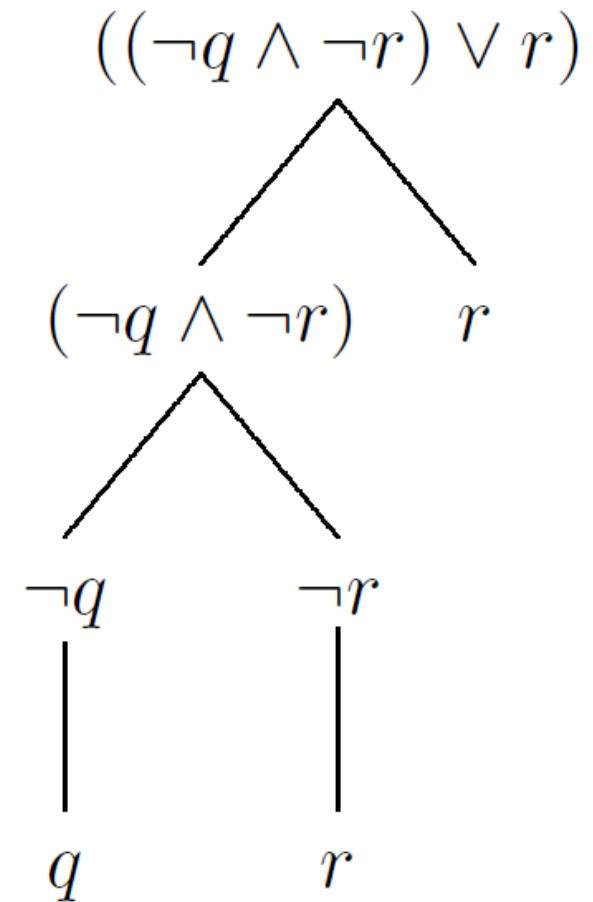
Then ...

...

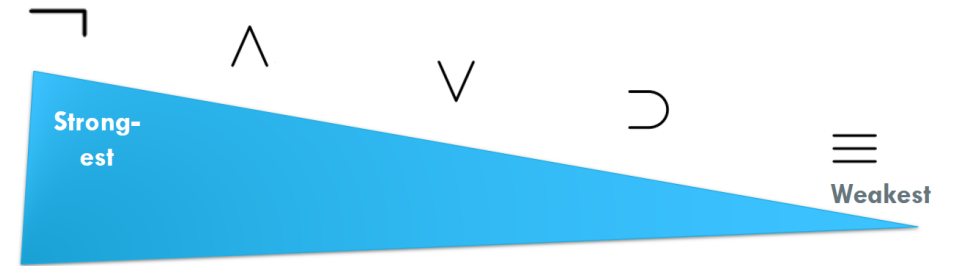
The leaves are atomic formulae

Exercise

Give the construction tree of the following formula:



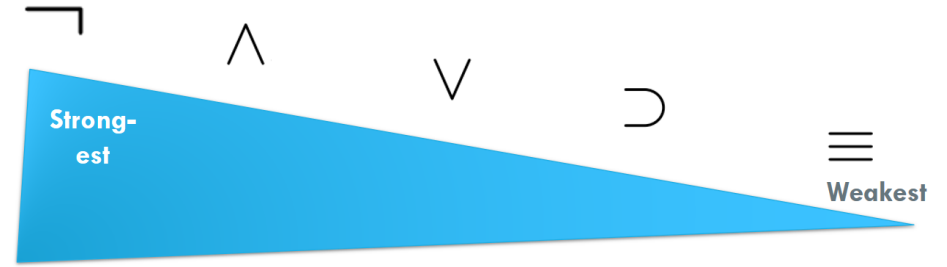
Exercises



A) Construction trees

1. Give the construction tree of *formula* of $\neg q \vee p \wedge q \supset r$!
2. Give the construction tree of *formula* of $\neg p \supset q \equiv \neg q \supset p$!
3. Give the construction tree of *formula* of $\neg(p \wedge q) \equiv \neg p \vee \neg q$!
4. Give the construction tree of *formula* of $p \wedge (q \vee r) \equiv p \wedge q \vee p \wedge r$!

Exercises



5. Give the construction tree of formula $((p \supset q) \supset p) \supset q$!
6. Give the construction tree of formula $\neg\neg(p \supset (r \wedge q))$!
7. Give the construction tree of formula $\neg(\neg(\neg p \supset \neg q) \supset (p \vee \neg\neg q))$!
8. Give the construction tree of formula $\neg(\neg(\neg p \equiv q) \equiv \neg r)$!
9. Give the construction tree of formula $(\neg(p \vee q) \equiv (\neg p \wedge \neg q))$!
10. Give the construction tree of formula $((p \supset (q \supset r)) \equiv (\neg(\neg p \vee q) \vee r))$!

Exercises — Construction trees

If surrounded by two occurrences of $\wedge$ or $\vee$, the operand associates with the operator to the left.

$$p \wedge q \wedge r \rightarrow ((p \wedge q) \wedge r)$$

$$p \vee q \vee r \rightarrow ((p \vee q) \vee r)$$

If surrounded by two occurrences of $\supset$ or $\equiv$, the operand associates with the operator to the right.

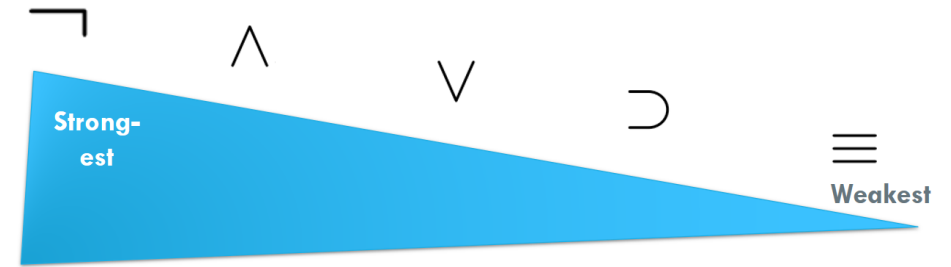
$$p \supset q \supset r \rightarrow (p \supset (q \supset r))$$

$$p \equiv q \equiv r \rightarrow (p \equiv (q \equiv r))$$

- (11)
- (a) $\neg\neg\neg(X \supset Y)$
 - (b) $(X \supset Y) \vee (X \supset Y \wedge Z)$
 - (c) $X \vee Y \supset \neg Z \wedge \neg(X \supset \neg Z)$
 - (d) $(X \supset Y) \wedge (Y \supset Z) \supset \neg X \vee Z$
 - (e) $(X \supset Y \wedge \neg Z) \supset (\neg X \supset \neg Y \wedge Z)$
 - (f) $\neg(X \vee Z \supset \neg Y) \supset \neg X \vee (\neg Y \wedge Z)$

$$\neg(p \wedge q \vee r \vee q \vee p \supset q \vee r \vee q \vee r \vee q \supset p \supset r \equiv q)$$

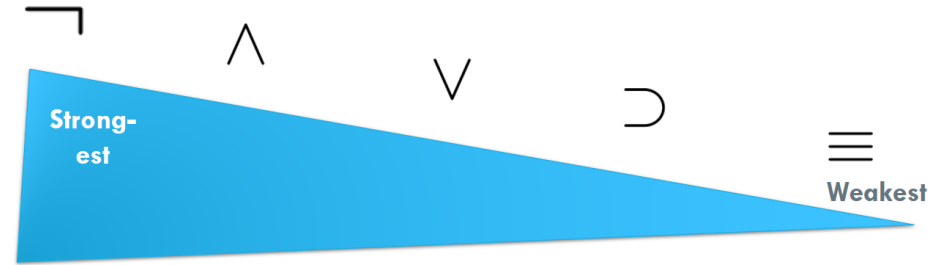
Exercises



B) Subformulae

10. Which of the following is a subformula of the formula $\neg q \vee p \wedge q \supset r$?
A. $q \vee p$ B. $p \wedge q$ C. $q \supset r$ D. $q \vee p \wedge q$ E. $\neg q$
11. Which of the following is a subformula of the formula $\neg p \supset q \equiv \neg q \supset p$?
A. $\neg p$ B. $q \supset p$ C. $\neg p \supset q$ D. $q \equiv \neg q$ E. $\neg p \supset q \equiv \neg q$
12. Which of the following is a subformula of the formula $\neg(p \wedge q) \equiv \neg p \vee \neg q$?
A. $q \wedge p$ B. $p \vee \neg q$ C. $p \wedge q$ D. $(p \wedge q) \equiv \neg p$ E. $\neg q$
13. Which of the following is a subformula of the formula $p \wedge (q \vee r) \equiv p \wedge q \vee p \wedge r$?
A. $p \wedge r$ B. $p \wedge (q \vee r)$ C. $(q \vee r) \equiv p$ D. $q \vee p$ E. $q \wedge p$
14. Which of the following is a subformula of the formula $p \vee q \wedge r \equiv (p \vee q) \wedge (p \vee r)$?
A. $r \equiv p$ B. $q \wedge r$ C. $p \vee q$ D. $q \wedge r \equiv (p \vee q)$ E. $r \equiv (p \vee q)$

Exercises



22. Give the direct subformulae of formula $\neg(\neg(\neg p \supset \neg q) \supset (p \vee \neg\neg q))!$
23. Give all the subformulae of formula $\neg(\neg(\neg p \supset \neg q) \supset (p \vee \neg\neg q))!$
24. Give the direct subformulae of formula $\neg(\neg(\neg p \equiv q) \equiv \neg r)!$
25. Give all the subformulae of formula $\neg(\neg(\neg p \equiv q) \equiv \neg r)!$
26. Give the direct subformulae of formula $((p \supset (q \supset r)) \equiv (\neg(\neg p \vee q) \vee r))!$
27. Give all the subformulae of formula $((p \supset (q \supset r)) \equiv (\neg(\neg p \vee q) \vee r))!$

Direct subformulae

16. Give the direct subformulae of formula $((\neg q \wedge \neg r) \vee r)!$
17. Give all the subformulae of formula $((\neg q \wedge \neg r) \vee r)!$
18. Give the direct subformulae of formula $((p \supset q) \supset p) \supset q!$
19. Give all the subformulae of formula $((p \supset q) \supset p) \supset q!$